Machine Learning

Course summary — Lectures 1 to 5

The whole course on three pages: what each chapter says, the formulas it lives on, and the questions it gets asked.

Chapter 1 — What does it mean to learn

Rules vs learning: a rule is written by hand and breaks on novelty; learning fixes a family of functions on data.

- Three forms of experience: supervised (labels), unsupervised (structure), self-supervised (labels manufactured from the input itself).
- Four ingredients of a learning system: data, a model family (hypothesis space), a loss, an optimisation procedure.
- Generalisation is the goal; shortcuts are patterns that correlate with the label but carry no signal (the dog-vs-grass trap).
- Uncertainty: entropy $H(p) = -\sum p \log p$, cross-entropy $H(p, q) = -\sum p \log q$, KL divergence $D_{\mathrm{KL}}(p\|q) = H(p, q) - H(p)$; cross-entropy is entropy plus the KL penalty for using q instead of p.

Chapter 2 — From representations to decisions

- Bayes decision rule: act to minimise expected loss; with 0-1 loss it is the argmax of the posterior. The Bayes error is the floor no model beats.
- Costs move the threshold: asymmetric losses turn a point prediction into a region; the reject option abstains inside a band.
- kNN: no training, decision by vote; bias grows with k, variance with $1/k$; the curse of dimensionality makes neighbours far and votes meaningless.
- ERM picks $\hat{f} = \arg\min \frac{1}{n} \sum \ell(f(x_i), y_i)$; training risk is optimistic (winner's curse). Cross-entropy as maximum likelihood: minimising it fits $p(y|x)$.
- Evaluation: train/validation/test, K-fold CV; never touch the test set before the final answer.

Chapter 3 — From fitting to generalisation

- The gap: training error minus true error; inductive bias is what makes small gaps possible (no free lunch).
- Capacity controls the trade-off: bias-variance decomposition $\mathbb{E}[(f(x) - y)^2] = \mathrm{bias}^2 + \mathrm{var} + \sigma^2$; the validation curve is U-shaped in capacity.
- Regularisation: ridge ($\lambda\|w\|_2^2$) shrinks, lasso ($\lambda\|w\|_1$) selects; early stopping is regularisation in time.
- Winner's curse: selecting on validation makes the reported number optimistic; nested CV gives an honest estimate.
- Leakage: any information of the test fold reaching training (scaling on all data, temporal splits ignored) inflates every score.

Chapter 4 — Discriminative and generative classification

- Discriminative: logistic regression $\sigma(w^\top x)$, softmax for multi-class; calibrated by construction when fitted by MLE.
- Generative: model $p(x|y)p(y)$; Gaussian classes give LDA (shared covariance $\Rightarrow$ linear) and QDA (per-class covariance $\Rightarrow$ quadratic).
- Naive Bayes: conditional independence given the class; smoothing handles unseen words; strong bias, low variance, works with little data.
- Learning curves decide between the two families: generative wins early, discriminative wins with data.
- Imbalance vs prior shift: with the score fixed, a new prior p' moves the threshold by $\log(p'/p)$; calibration (reliability diagram) before decisions.

Chapter 5 — Learning by combining models

- Voting: averaging m independent predictors divides variance by m; correlated committees barely help ($\mathrm{var}(\bar{f}) = \frac{1+\rho(m-1)}{m}\sigma^2$).
- Bagging: bootstrap samples train m trees; out-of-bag estimates are free validation; random forests add feature subsampling (m_{try}).
- Importance: MDI (in-training, biased to high-cardinality) vs permutation (held-out, honest).
- Gradient boosting: stagewise fits of the gradient; each leaf is a Newton step $w^* = -\frac{\sum g_i}{\sum h_i + \lambda}$, split gain $\frac{1}{2}\left[\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{(G_L+G_R)^2}{H_L+H_R+\lambda}\right] - \gamma$.
- Stacking: out-of-fold predictions feed a meta-learner; mixture of experts routes by input.

Key formulas

Cross-entropy $\ell = -\log p(y|x)$ • Bayes rule $a^* = \arg\min_a \sum_y p(y|x)\,\ell(a,y)$ • ridge/lasso penalties

Bias–variance $\text{bias}^2 + \text{var} + \sigma^2$ • committee variance $\frac{1+\rho(m-1)}{m}\sigma^2$ • Newton leaf $-\frac{\sum g_i}{\sum h_i+\lambda}$

Permutation importance: drop in score when a feature is shuffled on held-out data; MDI: total impurity reduction, in-training.

What the exam asks

Distinguish pairs (capacity vs variance, generative vs discriminative, ridge vs lasso, MDI vs permutation, bagging vs boosting); derive the small formulas (Bayes rule with costs, committee variance, logistic gradient, LDA weights, XGBoost leaf); read learning curves and reliability diagrams; design a leakage-free split; and always answer claim $\rightarrow$ mechanism $\rightarrow$ number $\rightarrow$ consequence.